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Evaluation of Multivalent Sialylated Polyglycerols for Resistance Induction in and Broad Antiviral Activity against Influenza A Viruses

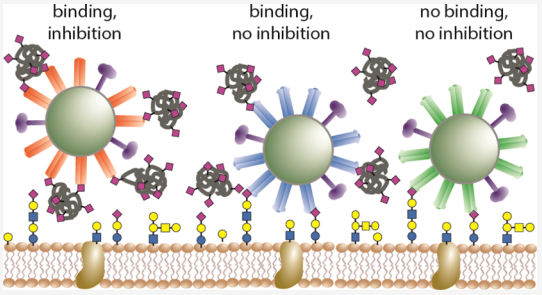
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ABSTRACT: The development of multivalent sialic acid-based inhibitors active against a variety of influenza A virus (IAV) strains has been hampered by high genetic and structural variability of the targeted viral hemagglutinin (HA). Here, we addressed this challenge by employing sialylated polyglycerols (PGs). Efficacy of prototypic PGs was restricted to a narrow spectrum of IAV strains. To understand this restriction, we selected IAV mutants resistant to a prototypic multivalent sialylated PG by serial passaging. Resistance mutations mapped to the receptor binding site of HA, which was accompanied by altered receptor binding profiles of mutant viruses as detected by glycan array analysis. Specifying the inhibitor functionalization to 2,6-α-sialyllactose (SL) and adjusting the linker yielded a rationally designed inhibitor covering an extended spectrum of inhibited IAV strains. These results highlight the importance of integrating virological data with chemical synthesis and structural data for the development of sialylated PGs toward broad anti-influenza compounds.



INTRODUCTION

Influenza virus causes acute respiratory infections, which lead annually to 3−5 million cases of severe illness and about 290.000 to 650.000 deaths worldwide.[1] Despite the availability of several classes of anti-influenza drugs, additional antiviral agents for effective treatment of influenza A virus (IAV) infections are needed urgently. Due to the rapid evolution of influenza virus, vaccination yields unsatisfactory protection in some years and resistance against antivirals can develop.[2] This has already occurred with M2 ion-channel inhibitors to which virtually all circulating strains have become resistant. Similar developments can be expected also for other classes of antivirals, as suggested by the emergence of neuraminidase inhibitor Oseltamivir-resistant seasonal IAV[3,4] and the clinical occurrence of variant viruses resistant to the new polymerase inhibitor baloxavir marboxil.[5,6] New classes of inhibitors targeting other viral proteins broaden the possibilities for combination therapies and limit viral escape.[7] The viral surface protein hemagglutinin (HA) is a promising drug target that could be applied in such combination therapies.[8]

A wide range of sialylated *N*- and *O*-glycans are expressed on cells of the airway epithelium.[9] Via HA, IAV attaches to glycans with terminal sialic acid (SA) on the surface of the host cell membrane. The receptor binding site (RBS) of the A/X31 virus HA (H3) is formed by the 130 loop (amino acids 134−138), the 220 loop (amino acids 224−228) to the sides, as well as the 190 helix (Glu190, Leu194) with His183 and Thr155 to the rear.[10] Tyr98 and Trp153 build the bottom of the binding grove. All amino acid positions are given in H3 numbering. Hydrogen bonds and van der Waals contacts form between the receptor SA and the RBS.[10] Amino acids in the vicinity of the RBS stabilize the binding pocket without interacting with the receptor directly.[11]

The individual HA-SA interactions are weak,[12] but IAV engages in multiple receptor binding events using several of its 300−400 trimers per virion,[13] thus stabilizing the attachment though high avidity.[14−16] While monovalent SA is an inefficient HA binder,[17−19] infection can be prevented by inhibiting virus binding competitively with multivalent compounds presenting SA units on their surface.[20−24] Several effects besides occupying the RBS contribute to successful inhibition, such as the chelate effect, statistical rebinding, secondary binding events, and steric shielding.[15,25]

Several sialylated multivalent constructs that compete with HA receptor binding have been tested so far. Some of the first multivalent systems were based on polyacrylate[23] or polyacrylamide[20−22] carrier backbones, which showed increased potency compared with monovalent versions of the compounds. Although they were useful for determining optimal

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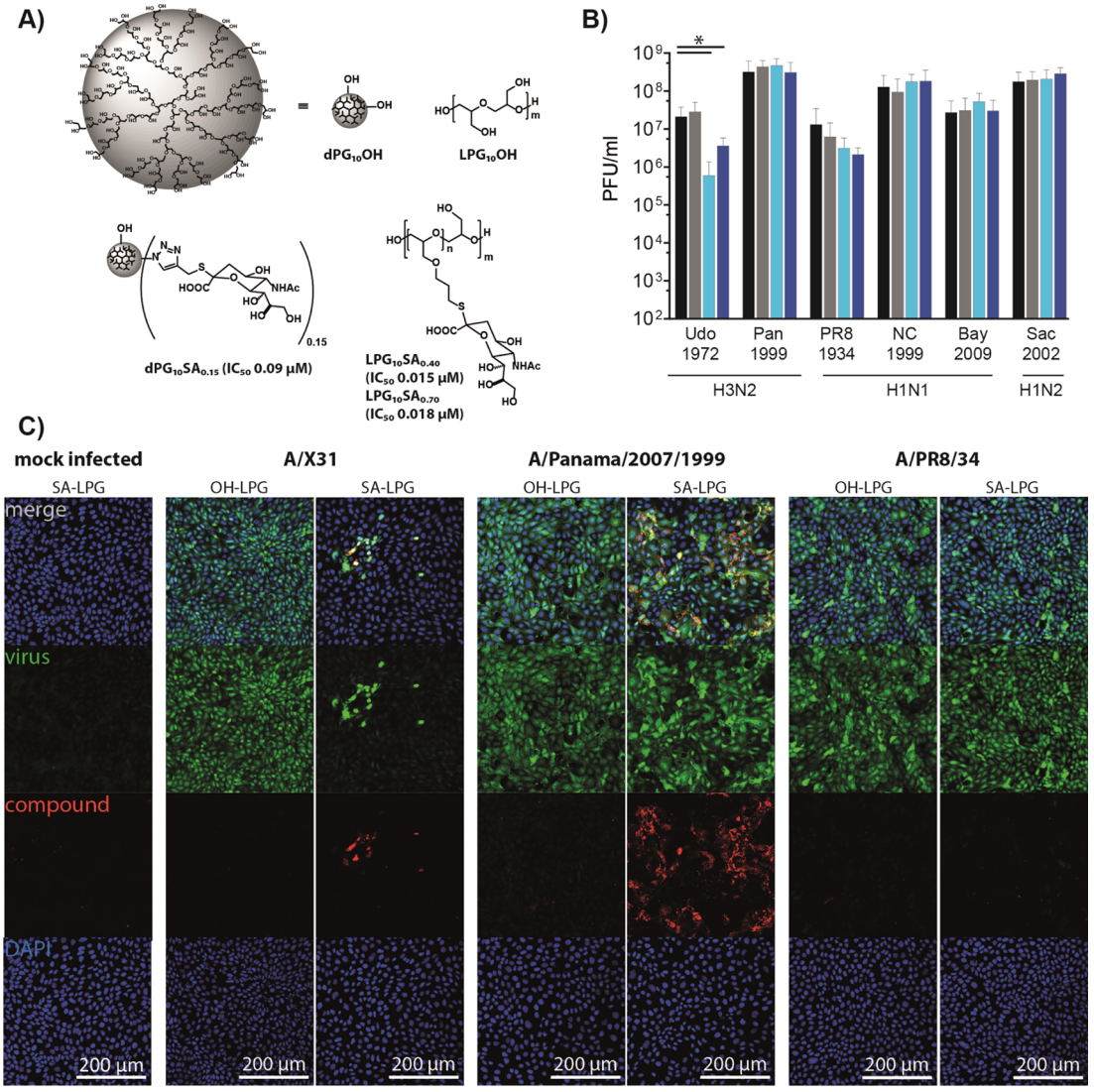


Figure 1. Effect of multivalent sialylated PGs on diverse IAV strains. (A) Structures of dendritic and linear PG polymers and multivalent sialosides, also including IC_{50} values toward influenza A/X31 virus in tissue culture, $dPG_{10}OH$, $LPG_{10}OH$, $dPG_{10}SA_{0.15}$, $LPG_{10}SA_{0.40}$, and $LPG_{10}SA_{0.70}$. (B) Inhibition of virus infection using SA-PGs. MDCK II cells were infected with influenza virus strains A/Udorn/1972 (Udo 1972), A/Panama/2007/1999 (Pan 1999), A/PR/8/1934 (PR8 1934), A/NewCaledonia/20/1999 (NC 1999), A/Bayern/63/2009pdm (Bay 2009), or A/Sachsen/1816/2002 (Sac 2002) at multiplicity of infection (MOI) 0.01 and treated with 100 μM [SA] of $LPG_{10}OH$ (gray), $LPG_{10}SA_{0.40}$ (blue), $LPG_{10}SA_{0.70}$ (dark blue), or left untreated (black) for 24 h. Each column represents the mean of three independent experiments. Error bars depict the standard deviation (SD). *P*-values were calculated using the Kruskal–Wallis test with Dunn's correction for multiple comparisons. * indicates a *p*-value <0.05. (C) Immunofluorescence staining of infected and PG-treated cells using Cy3-labeled PGs. MDCK II cells were infected with the indicated virus strain at MOI 0.01 for 45 min or mock-infected and treated with 5 μM [PG] of Cy3-$LPG_{10}SA_{0.40}$ or Cy3-$LPG_{10}OH$ for 24 h. Cells were fixed and prepared for immunofluorescence microscopy using suitable primary and secondary antibodies. Viral NP is shown in green color, Cy3-labeled PG is colored red, and cell nuclei were stained with DAPI (blue).

design parameters,[26] their high toxicity[27,28] and low biocompatibility[29] limit their applicability for *in vivo* studies. Sialylated polyamidoamine (PAMAM)-based scaffolds have also been widely studied,[30−33] but concerns regarding the toxicity have also been raised for this backbone.[34] Other sialylated materials aimed at improved biocompatibility have been successful at inhibiting influenza virus HA, such as chitosan,[35,36] liposomes,[37−40] gold nanoparticles,[41,42] phage capsids,[43] and polyglycerol (PG).[44,45] These studies have demonstrated that flexibility of the scaffold,[15,45] size of inhibitors,[42,44,46] and the linker between SA and the carrier backbone[32,43,47−49] can significantly influence the binding to IAV.

Among the different multivalent approaches, functionalized PGs have proven advantageous. They are water-soluble, highly biocompatible, and easy to functionalize because of several hydroxyl groups on the PG backbone.[50−52] Some oligoglycerol esters are even FDA-approved and have been used as food or pharmaceutical additives.[53] Sialylated PGs inhibit IAV attachment and therefore infection most likely by competing with the binding of SA to the RBS.[15,45] Especially the high flexibility is important because scaffold configuration is fundamental for effective inhibition.[44,46,54] On this matter, we recently synthesized and compared linear and dendritic PG scaffolds with an average molecular weight of 10 kDa to gain insights into the preferred scaffold geometry needed for effective inhibition of the widely studied influenza A/X31 virus

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(H3N2).[45] The optimized linear PG with 40% SA functionalization ($LPG_{10}SA_{0.40}$) proved to be much more potent than the optimized dendritic PG ($dPG_{10}SA_{0.15}$) and inhibited influenza virus A/X31 (H3N2) infection in nanomolar concentrations. This detailed investigation also revealed the optimum degree of sialylation as 40−70 and 15−20% on linear and dendritic PGs, respectively.[45] However, the potential for broad activity of these sialylated PG constructs against a variety of IAV strains has not been fully explored so far.

Broad activity is important because influenza viruses evolve and accumulate mutations, especially in the HA gene, resulting in antigenic drift.[55,56] In particular, amino acids in the RBS of HA are highly variable. For instance, the RBS sequences of H3N2 subtype viruses from 1968 and 2012 differ by around 48% identity on the amino acid level.[57] This high degree of variation not only explains antigenic drift, but also changes in the receptor binding profiles over time with an increased preference of the more recent H3N2 strains to bind to long, branched glycans.[58−60] Because receptor binding properties differ considerably between influenza virus strains, it is essential to test compounds raised against HA for broad activity,[23,31,33,61] a fact often neglected.[21,26,62−66]

Here, we report that previously described sialylated PG inhibited only a narrow range of tested H1N1 and H3N2 subtype IAV strains. To further elucidate the molecular determinants of resistance or susceptibility, we for the first time selected mutant IAV with resistance to a multivalent sialoside, namely, $LPG_{10}SA_{0.40}$, and investigated the developed mutations for their impact on receptor binding specificity by glycan array analysis. By adjusting functionalization and linker of the PGs, we designed a new multivalent inhibitor that is active against a variety of IAV H3N2 and H1N1 strains representing nearly 50 years of viral evolution. Here, we report that broad anti-influenza activity, previously not observed for other multivalent SA-conjugates,[30] can be achieved by integrating virological data with chemical synthesis and structural considerations in the design of improved inhibitory compounds.

RESULTS

Multivalent Sialylated PGs Inhibit Only Selected Strains of IAV. Previously, we demonstrated the inhibitory potential of several LPGs and dPGs against influenza virus A/X31 that are nontoxic *in vivo*.[45] We further evaluated three of these compounds, $LPG_{10}SA_{0.40}$, $LPG_{10}SA_{0.70}$, and $dPG_{10}SA_{0.15}$ (Figure 1A) in a cell-based therapeutic approach. These analyses revealed potent inhibition of influenza A/X31 virus (H3N2) for the three PGs with IC_{50} values of 0.015, 0.018, and 0.25 μM of PG (corresponding to 0.9, 1.8, and 6.5 μM of SA when considering the number of ligands per PG), respectively (Supporting Information Figure S1). For comparison, we found the IC_{50} value of Oseltamivir to be 2.74 μM under these conditions, which matches literature data for cell-based assays.[67−69] To evaluate whether the PGs can target diverse IAV strains, we extended the panel of virus strains to include two additional H3N2 viruses, three H1N1 viruses, as well as an H1N2 virus (Figure 1B). As a result, we found that in this panel of viruses, only the A/Udorn/1972 (H3N2) virus was susceptible to treatment with 100 μM [SA] of linear PGs. However, propagation of A/Panama/2007/1999 (H3N2) as well as the H1 subtype strains A/PR/8/34, A/NewCaledonia/20/1999, A/Bayern/63/2009, and A/Sachsen/1816/2002 (H1N2), which are significantly diverged from H3 viruses in their HA amino acid sequence, was not inhibited by the tested SA-functionalized LPGs. Results from samples treated with control compound without sialylation ($LPG_{10}OH$) were comparable to untreated samples. This finding indicated that even subtle differences in the HA RBS may alleviate susceptibility to the tested PG compounds.

To address the question whether unresponsiveness of a virus strain toward a given compound is due to a lack of HA binding, we prepared fluorescently Cy3-labeled $LPG_{10}SA_{0.40}$ as well as nonfunctionalized $LPG_{10}OH$ (refer to the Supporting Information for synthesis details and spectral data) to visualize compound localization and infection inhibition in cell culture (Figure 1C). In this assay, MDCK II cells were grown on glass cover slips, infected with different IAV strains (MOI = 0.01) and incubated in the presence of 5 μM [PG] of Cy3-$LPG_{10}SA_{0.40}$ or Cy3-$LPG_{10}OH$. Subsequently, cells were fixed and stained for viral antigen as well as for nuclei by immunofluorescence microscopy (Figure 1C). No compound was detected on slides with mock infected cells, confirming that the PGs did not bind to MDCK II cells. The control compound without any SA residues, Cy3-$LPG_{10}OH$, was not detected either, indicating that PGs did not bind nonspecifically. Viral infection was visualized by staining the viral nucleoprotein (NP), and thus, infected cells are seen in green. In the absence of sialylated PG, A/X31 virus infected the whole layer of cells. In contrast, the presence of Cy3-$LPG_{10}SA_{0.40}$ reduced infection to a few isolated infected cells. Cy3-$LPG_{10}SA_{0.40}$ was detected only on infected cells. In contrast, the A/Panama/2007/1999 (H3N2) and A/PR8/34 (H1N1) viruses infected the whole layer of cells both in the presence and in the absence of Cy3-$LPG_{10}SA_{0.40}$, which was expected because both viruses were not inhibited by SA-LPGs (Figure 1B). Unexpectedly, however, Cy3-$LPG_{10}SA_{0.40}$ was codetected on many cells infected with A/Panama/2007/1999-, but not in A/PR/8/34-infected cells. It is therefore likely that even subtle differences occurring within the HA RBS of the same subtype can render inhibition by SA-functionalized PGs ineffective by reducing the binding affinity for SA, without necessarily abolishing SA binding. In contrast, the large differences between H1N1 and H3N2 HAs[70] may explain the lack of compound binding and lack of inhibition of A/PR/8/34 (H1N1) virus.

Resistance against Sialylated PG Develops Rapidly and Mutations Map to the RBS. In order to analyze which viral components determine susceptibility or resistance to LPG-based sialylated compounds and to determine the threshold of resistance, we generated A/X31 virus variants that were resistant against $LPG_{10}SA_{0.40}$ by serial passaging (Figure 2). Cells were infected with influenza A/X31 virus at an MOI of 0.01 in triplicate wells and treated with an inhibitor. After 24 h, the supernatants of all wells were titrated using plaque assay and used to infect the subsequent passage with the same MOI 0.01. The initial inhibitor concentration was selected around the IC_{50} value and escalated with the increasing passage number. Already at passage 2, the virus grew to 1 to 2 orders of magnitude higher titers with the initial compound concentration of 1.67 nM [PG] (0.1 μM [SA]). The compound concentration was increased gradually to 1666.7 nM [PG] at passage 9. A concentration of 1666.7 nM [PG] of $LPG_{10}SA_{0.40}$ did not substantially reduce virus titers by passage 10 (Figure 2). For comparison, A/X31 virus was passaged with a similarly stringent escalating concentration of Oseltamivir. At passage 10, the propagation of two of the three

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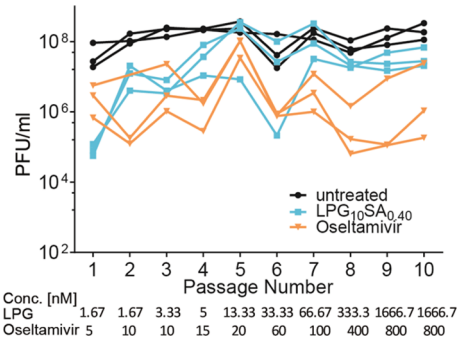


Figure 2. Serial passaging of influenza virus strain A/X31 with $LPG_{10}SA_{0.40}$ (blue), Oseltamivir (orange), or mock treatment (black). At the initial passage, MDCK II cells were infected with A/X31 virus at MOI 0.01 and treated with either $LPG_{10}SA_{0.40}$ or Oseltamivir at the indicated concentration or left untreated for 24 h. Supernatants were titrated using plaque assay. Each subsequent passage was infected with a supernatant from the previous passage at MOI 0.01 and treated with either $LPG_{10}SA_{0.40}$ or Oseltamivir at the indicated concentration or left untreated for 24 h. Each replicate is displayed separately to demonstrate the unique differences in the progressing adaption.

replicates was still inhibited by Oseltamivir by more than 1 order of magnitude at a compound concentration 160-fold of the initial concentration. Differences in virus titers between the three replicates of $LPG_{10}SA_{0.40}$ or Oseltamivir treated viruses were observed at most passages, indicating that adaption did not proceed identically for individual replicates.

Viruses from passages 4, 7, and 10 and the initial stock virus were subjected to next-generation sequencing (NGS). All mutations were the result of single nucleotide substitutions. All mutations detected in HA in the passaged viruses evolved independently from each other in the triplicate samples and were compared with the initial A/X31 virus stock, as listed in Table 1. All HA mutations are given in H3 numbering. $LPG_{10}SA_{0.40}$-treated viruses had mutations mostly in HA. Mutation G135E was detected in $LPG_{10}SA_{0.40}$-treated replicate 1 (L1) at passage 4 with an abundance of 89.8%, which solidified to 98.4% at passage 10 with no other mutations. In replicate 2 (L2), mutations K140I and K140N were detected at passage 4 in 60.9 and 29.5% of the sequences, respectively. The K140N mutation proved dominant in 97.3−99.9% of the sequences between passages 7 and 10. Replicate 3 (L3) had mutations G144D and R224M at 38.4 and 43.9%, respectively, at passage 4, though only the former prevailed and reached 99.6% at passage 10. Additionally, P99F occurred at 20.6% in L3 at passage 7 and solidified to 94.8% at passage 10. This coincides with an increase in virus titer of L3 starting from passage 7. No other mutations were detected in L3. None of the $LPG_{10}SA_{0.40}$-passaged viruses carried mutations in the NA gene. Clonal viruses from passage 10 were plaque-purified and the HA sequence including the mutations confirmed by Sanger sequencing, resulting in mutants L1 (G135E), L2 (K140N), and L3 (P99F/G144D). As anticipated, the mutations are located in or close to the RBS of HA. Samples passaged without treatment had no mutations, suggesting that the A/X31 genome is stable upon passage in MDCK cells.

Oseltamivir-treated viruses had several mutations in HA. No mutation in NA was detected, despite Oseltamivir being an NA inhibitor. However, HA mutations decreasing receptor affinity and conferring resistance to NA inhibitors have been selected *in vitro* before.[71,72] One of the LPG passaged viruses and all three Oseltamivir-selected variants showed single additional mutations in their PB2 genes, which are listed in Supporting Information Table S1, but have not been investigated further.

Resistance Mutations Confer Cross-Resistance against Other Sialylated PGs. The mutations did not cause a loss of fitness in the mutant viruses, as they grew to similar if not slightly higher titers compared to the parental virus (Figure 3A). Virus titers of the $LPG_{10}SA_{0.40}$-selected mutant viruses were not reduced by $LPG_{10}SA_{0.40}$, confirming resistance to this compound.

The $LPG_{10}SA_{0.40}$ used for selection of resistant variants is only one representative of several other multivalent inhibitory PG molecules that differ in the topological presentation of their functionalized SA groups. To understand the significance of the topology by which SA is presented to the viral HA for resistance, we investigated inhibition of the selected mutant viruses by two classes of SA-functionalized inhibitors of A/X31: first, by linear $LPG_{10}SA_{0.70}$ and the dendritic $dPG_{20}SA_{0.15}$, previously described potent PG-based inhibitors of A/X31

Table 1. Variant Frequency of HA Mutations in the Three Replicates at Passage 4, 7, and 10 Selected Using Oseltamivir or $LPG_{10}SA_{0.40}$ Compared with WT X31 Virus as Determined by NGS

	passage 0	passage 4			passage 7			passage 10		
		replicate			replicate			replicate		
HA mutation[a]	reference[b] (%)	1 (%)	2 (%)	3 (%)	1 (%)	2 (%)	3 (%)	1 (%)	2 (%)	3 (%)
					LPG[c]					
P99F	0.1	0.0	0.1	0.0	0.0	20.6	0.0	0.0	94.8	0.1
G135E	0.1	89.8	0.0	0.0	98.5	0.0	0.0	98.4	0.0	0.0
K140I	0.0	0.0	0.0	60.9	0.0	0.0	1.9	0.0	0.0	0.0
K140N	0.0	0.0	0.1	29.5	0.0	0.0	97.3	0.0	0.0	99.0
G144D	0.0	0.2	38.4	0.0	0.0	98.6	0.0	0.0	99.6	0.0
R224M	0.3	0.1	43.9	0.0	0.4	74.3	0.3	0.5	2.3	0.3
					Oseltamivir					
G135E	0.1	0.0	0.0	0.0	0.1	68.4	0.1	0.0	98.5	0.1
N188D	0.1	35.5	82.0	0.1	50.9	99.2	0.1	26.7	99.4	0.0
S228G	0.0	0.0	0.0	0.6	0.0	0.0	10.0	0.0	0.1	90.4
N296T	0.5	35.1	79.5	1.6	47.5	99.4	0.1	28.6	99.5	0.1
A495T	0.1	0.4	0.9	41.2	0.3	0.0	87.8	0.0	0.0	6.7

[a]According to H3 HA numbering. [b]Reference corresponds to the sequence of the initial virus used to infect passage 1. [c]$LPG_{10}SA_{0.40}$.

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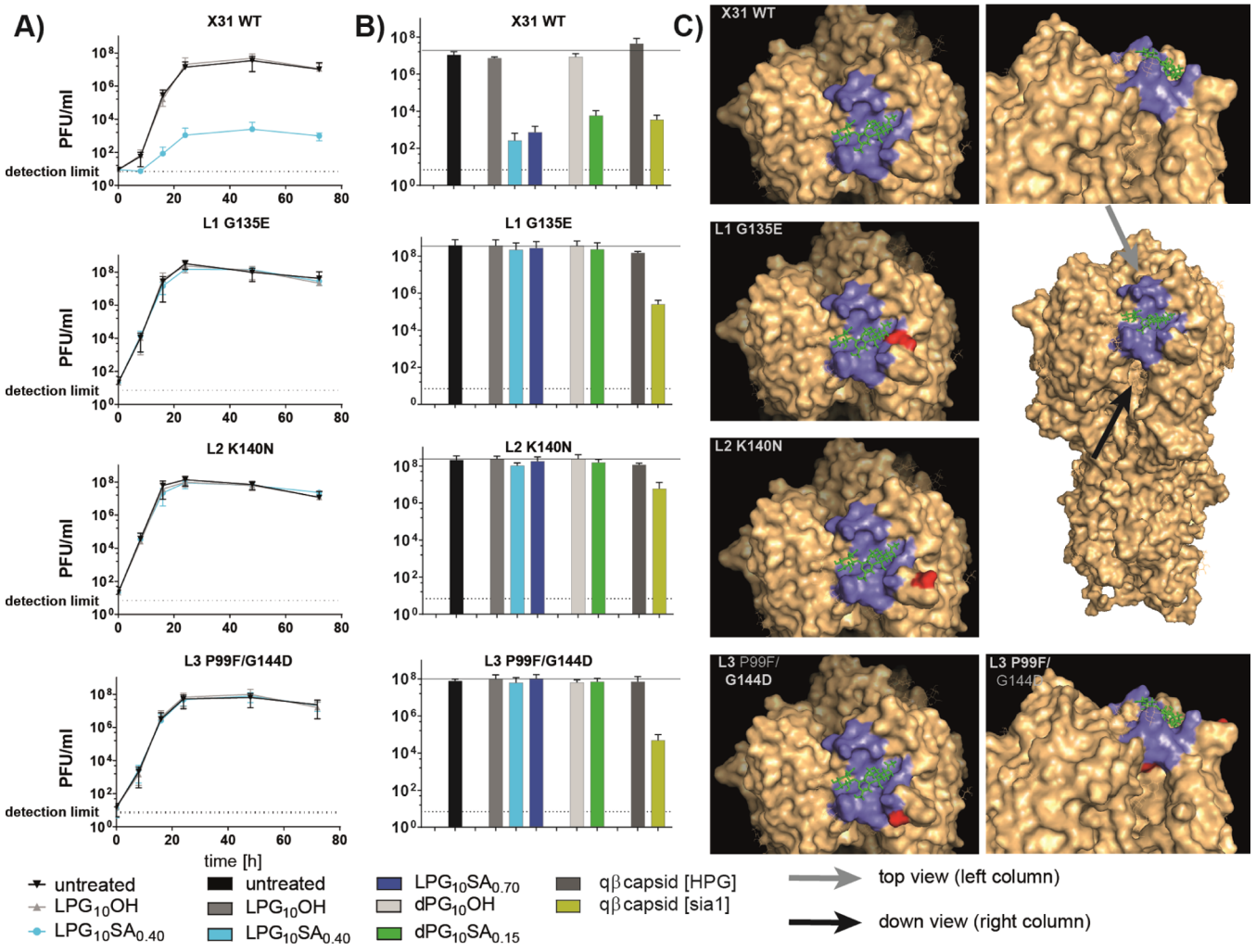


Figure 3. Characterization of A/X31 WT and selected mutant viruses for inhibition by SA-functionalized compounds. (A) Growth curve analysis of A/X31 WT and mutant viruses on MDCK II cells in the presence of $LPG_{10}SA_{0.40}$ (blue) or $LPG_{10}OH$ (gray) being almost identical to without treatment (black). MDCK II cells were infected at MOI 0.01 and treated with the indicated compound at 1.67 μM [PG] or mock-treated; supernatants were titrated using plaque assay at the indicated time periods. Data represent three independent experiments in duplicates and error bars depict the SD. (B) Cross-resistance of mutant viruses against 100 μM [SA] of PG-SA conjugates: $dPG_{20}SA_{0.15}$ (green), $dPG_{10}OH$ (light gray), $LPG_{10}SA_{0.70}$ (dark blue), $LPG_{10}SA_{0.40}$ treatment (blue), $LPG_{10}OH$ treatment (gray), or 7 nM [capsid] of phage capsid conjugates: Qβ capsid [sia1] (olive green) refers to sialylated capsid with a monoethylene glycol linker, whereas Qβ capsid [HPG] (dark gray) refers to capsid without conjugation[43] or without treatment (black). MDCK II cells were infected at MOI 0.01 and treated or mock-treated for 24 h. Supernatants were titrated using plaque assay. Each column represents the mean of three independent experiments in duplicates, and error bars depict the SD. (C) Crystal structure of the HA protein of A/X31 (PDB, 1HGG)[73] with RBS (purple), SA receptor analogue (green), and mutated amino acids (red) generated using PyMOL.[74] To enable visualization of mutation P99F in L3, a close-up at a different angle on HA is given in the lower right field. The arrows indicate the approximate angle the RBS close-ups are viewed with, the gray arrow corresponding to top view (left column) and the black arrow corresponding to down view (right column, only WT and L3).

(Figure 1).[45] Second, SA-functionalized capsids derived from the Qβ bacteriophage were recently shown to be promising multivalent inhibitors of IAV HA by Lauster et al.[43] This analysis showed that the three selected mutant viruses L1, L2, and L3 were completely cross-resistant to the two other PG-based compounds $LPG_{10}SA_{0.70}$ and $dPG_{20}SA_{0.15}$ (Figure 3B). In contrast, SA-functionalized phage capsid Qβ[Sia1][43] reduced viral titers of the mutant viruses by one to three orders of magnitude, although the inhibition was not as pronounced as determined for the parental A/X31 in parallel (Figure 3B). These findings indicated that all three PG compounds target tightly overlapping structural elements of the HA RBS for inhibition, whereas the phage capsid compound appears to interact with the HA RBS in a different or at least more robust manner, despite all compounds being functionalized with the same SA.

Receptor Binding of IAV Mutants is Altered by Mutations in HA RBS. Single-virus force measurements using atomic force microscopy (AFM) were employed to compare the interaction of a prototypic receptor analogue with WT and single (L1) or double (L3) mutant viruses. For this purpose, 6′-sialyllactose (SL)-biotin receptors were spotted on a poly(ethylene glycol) (PEG)-coated glass layer via streptavidin−biotin interaction, and virions were covalently attached to AFM cantilevers by bifunctional PEG-crosslinkers. The cantilevers were brought into proximity of the coated glass layer and then retracted again. Rupture forces obtained by retraction of the cantilever provide a measure for the bond strength between virion and receptor (Figure 4). The dynamic

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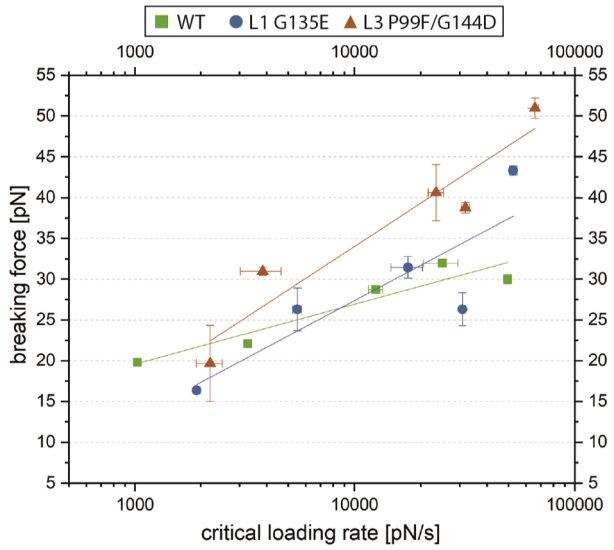


Figure 4. Dynamic force spectra of single-virus receptor interactions as determined by AFM. The SL receptor was spotted on a PEG-coated glass layer, and influenza virions were attached to AFM cantilevers. Rupture forces were measured by retraction of the cantilever from the receptor-coated surface. Most probable rupture forces were plotted against critical loading rates, and by application of the Bell–Evans model[75] (straight lines), kinetic parameters were extracted. Error bars represent standard error calculated from the variance–covariance matrix. For each data point, $46 \leq n \geq 123$.

force spectra were fitted as described in the material section and revealed that the kinetic off rate constant k_{off} was around 20- to 30-fold higher for mutant viruses (L3: k_{off} = 15.46 s^{-1}, L1: k_{off} = 19.93 s^{-1}) than for WT virus (k_{off} = 0.68 s^{-1}). In other words, the average bond lifetime τ_{off} was around 20- to 30-fold higher for WT virus than for mutant viruses (Table 2).

Table 2. Rupture Length x_b (Distance Over Which the Force Acts Until Bond Rupture[77]), Kinetic Off Rate k_{off}, and Average Bond Lifetime τ_{Off} as Calculated from Fitting the Data in Figure 4 to eq 1[a]

	x_b (nm)	k_{off} (s^{-1})	τ_{off} (s)
X31 WT	1.27 ± 0.25	0.68 ± 0.99	1.475
L1 G135E	0.65 ± 0.23	19.93 ± 26.89	0.050
L3 P99F/G144D	0.53 ± 0.10	15.46 ± 11.15	0.065

[a]Errors correspond to the standard error calculated from the variance–covariance matrix.

Moreover, the differences of the rupture length clearly point to different free energy landscapes between WT virus and mutant viruses. Here, the free energy landscape determines bond kinetics, as shown in detail previously.[75,76] Although the kinetic off rate k_{off} is less well defined as indicated by the larger error, the differences are in line with the abovementioned data obtained by independent approaches. Thus, although transient binding of mutants was still observed, their stability of binding to the receptor was reduced substantially.

Glycan array analysis has been widely used to investigate influenza HA receptor binding specificities.[57,58,78] To compare the binding specificities of wild-type and the $LPG_{10}SA_{0.40}$-resistant mutant viruses, we employed two different glycan arrays. On a low-density array, 25 glycans were spotted, including 14 sialylated glycans, four of which contain terminal 6′-linked *N*-acetylneuraminic acid (Neu5Ac), which is the canonical receptor of human influenza viruses.[79] As expected, WT virus bound to the four human-type receptor determinants with terminal 6′-Neu5Ac spotted on the array with high binding intensity (Figure 5, array 1). Some glycans with terminal 3′-Neu5Ac, the typical receptor of avian IAV,[79] were also bound by WT virus, which is not surprising, as A/X31 virus was grown as a prototypic vaccine strain in chicken eggs, which express this glycan abundantly.[80] L1 and L2 bound to three of the glycans with terminal 6′-Neu5Ac with around 100-fold reduced binding intensity compared with wild type and did not show binding to any other glycan presented on the array. L3 did not bind any of the glycans presented on the array. This indicated that the mutant viruses had lost detectable binding to the typical human-type sialylated receptors.

To elucidate the spectrum of glycans bound by the mutant viruses, we employed a second comprehensive array featuring 100 glycans (Figure 5, array 2). Again, WT virus bound to several typical IAV receptor determinants, that is, glycans with terminal 6′-Neu5Ac or 3′-Neu5Ac,[78,81] whereas binding of the viral mutants to any glycan with terminal 6′-Neu5Ac was rarely observed. Instead, L1 and L2 bound several distinct glycans with α2,8-linked SA and with *N*-glycolylneuraminic acid (Neu5Gc). L3 on the other hand bound some fucosylated glycans and glycans with α2,3-linked SA. All three mutant viruses bound glycans with terminal *N*-acetylclucosamine (GlcNAc), and binding to terminal galactose was also observed, especially for L2. All viral mutants bound strongly to all six aminoglycosides presented on the array, whereas WT virus recognized sisomicin sulfate, neomycin sulfate and, with weak binding intensity, also gentamicin sulfate and kanamycin sulfate. Taken together, these data indicate that the mutant viruses lost the high affinity binding to the typical IAV glycan receptors, while maintaining or acquiring the ability to bind different glycan receptors of the vast variety expressed on MDCK cells.[82−84]

Because the mutant viruses were resistant to inhibition by SA-functionalized PG but not SA-functionalized phage capsids (Figure 3B), we investigated by indirect immunofluorescence analysis whether Cy3-labeled $LPG_{10}SA_{0.40}$ would still be retained by viral HA expressed on infected cells (Figure 6). Similar to the case with A/Panama/2007/1999 (Figure 1C), Cy3-$LPG_{10}SA_{0.40}$ decorated cells infected with A/X31 WT or mutant viruses (Figure 6). Cy3-labeled $LPG_{10}OH$ control compound did not bind and was thus washed off when preparing the slides (Supporting Information Figure S4). Therefore, we hypothesize that the mutant viruses have strongly decreased attachment to $LPG_{10}SA_{0.40}$, but do bind to an altered spectrum of glycans with terminal SA present on MDCK cells, allowing them to propagate efficiently.

Synthesis of Multivalent Sialyllactosides Based on Linear Polyglycerol. Data so far indicated that recognition of SA-functionalized PGs to viral HA is very sensitive to variations and that even a single amino acid change in HA may render an PG-SA-based inhibitor ineffective. Moreover, antiviral activity is strongly affected by the specific presentation of the SA on a multivalent scaffold (PGs and phage capsids). Small changes in PG configuration such as SA density and PG flexibility were not sufficient to extend inhibition to viruses of the same H3N2 subtype with amino acid variations in the HA RBS or to different subtypes (Figures 1B and 3B).

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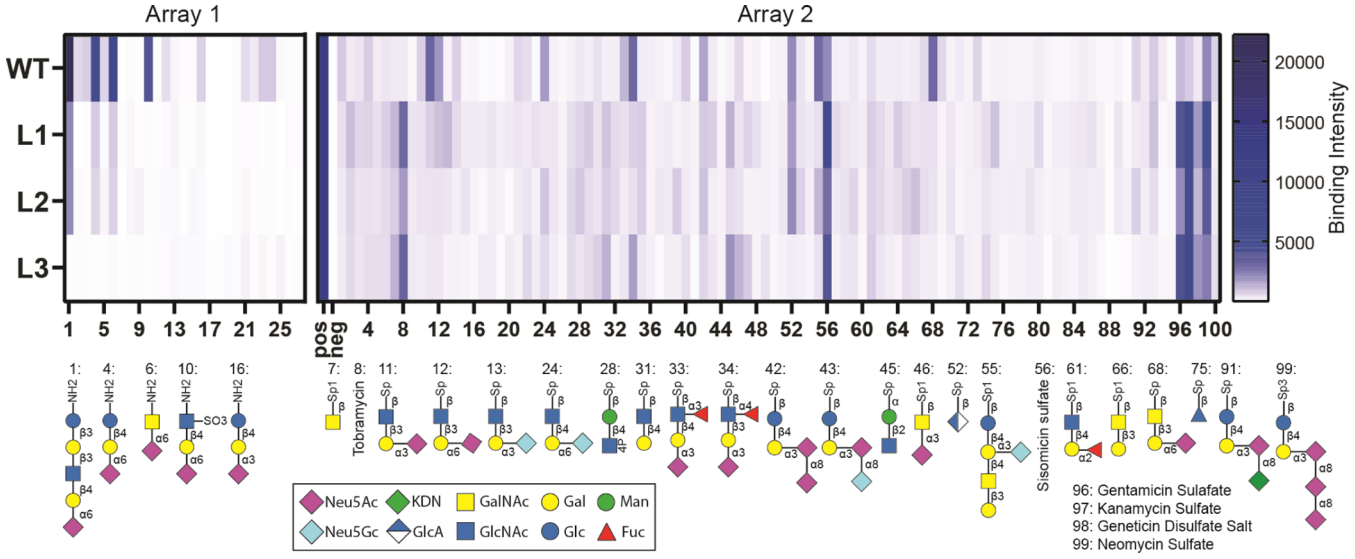


Figure 5. Binding profile of WT and mutant viruses on a low-density glycan array (array 1) and a commercial glycan array (array 2). Selected glycans that are bound by the virus are displayed below. The full list of glycans on the arrays is given in Supporting Information Tables S2 and S3. Binding intensity was calculated from the mean of three independent experiments. Individual slides were normalized to the positive control.

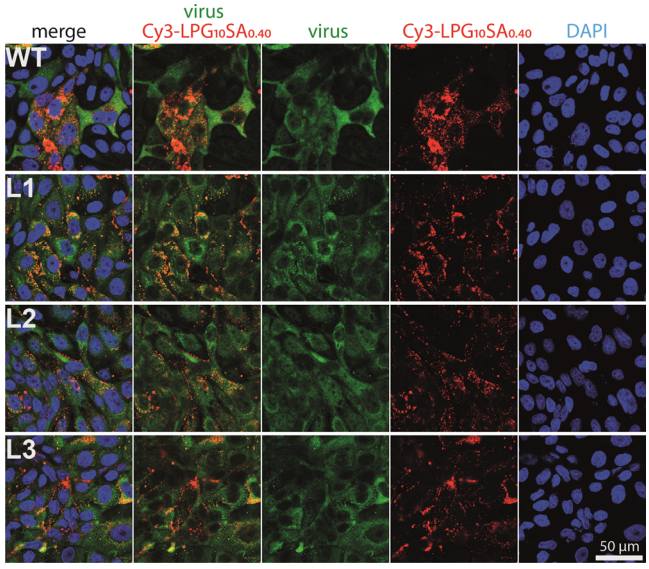


Figure 6. Immunofluorescence staining of infected and PG-treated cells. MDCK II cells were infected with MOI 0.01 for 45 min and incubated with 10 μM of compound for 24 h. Viral NS1 is colored green, Cy3-labeled $LPG_{10}SA_{0.40}$ is colored red, and cell nuclei are stained with DAPI (blue). Figure S4 shows staining with Cy3-labeled unsialylated control compound $LPG_{10}OH$.

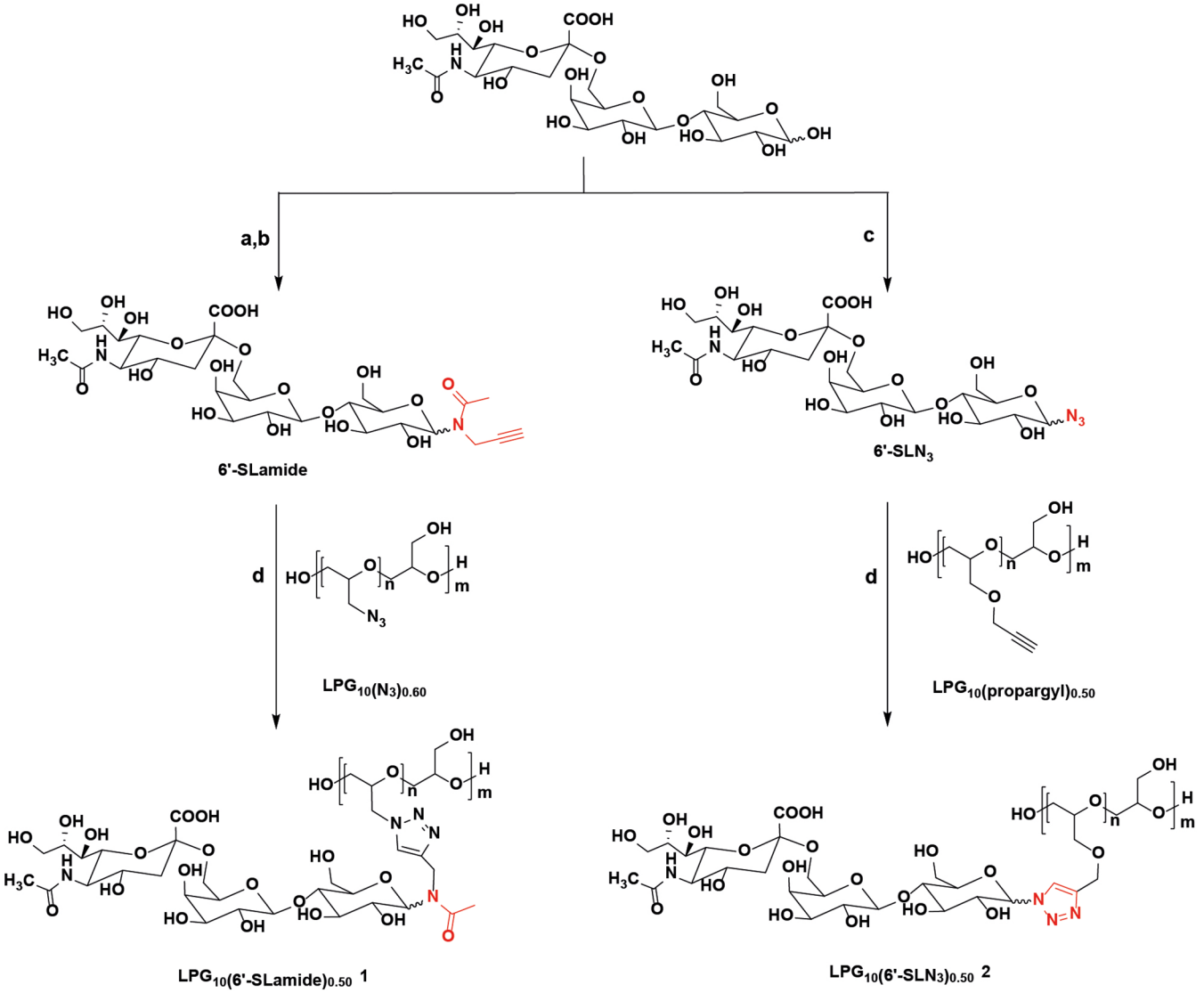
Toward our goal to render PG-based IAV inhibitors less susceptible toward HA amino acid variations and thereby to potentially broaden the number of susceptible strains, we sought to change the functionalization of and the presentation by the PGs. SA is the terminal sugar on highly diverse sugar structures,[9] and the underlying glycans are known to impact influenza virus binding.[85,86] While binding to α-2,6-sialyllactose (SL) was reduced in mutant viruses L1 and L2 compared to WT virus and nearly abolished by L3 (Figure 5, array 1), it was still one of the few glycans that was at least partly shared between wild-type and mutant viruses. Furthermore, we rationalized that SL serves as a structure bound by a variety of viruses on a variety of glycan arrays.[78] Hence, we synthesized PGs functionalized with sialyllactose (SL) in 6′ configuration to aim at a broader spectrum of inhibited influenza viruses. Because presentation of the SA appeared to be important for inhibition (Figure 3B), we prepared two SL-functionalized LPG with different linkers in order to be able to investigate two slightly different presentations of the sugar groups on the PGs: first, we prepared the α-2,6-sialyllactose azide (6′-SLN_3) by a method reported earlier.[87] Another new derivative, α-2,6-sialyllactosylamide (6′-SLamide), was prepared first by amination at the anomeric position with propargyl amine and second by N-acetylation with acetic anhydride. Both derivatives were obtained in yields above 90%. Using Cu-assisted click chemistry, 6′-SLN_3 and 6′-SLamide were conjugated with 10 kDa-based $LPG(propargyl)_{DF}$ and $LPG(N_3)_{DF}$, respectively, with a degree of functionalization (DF) around 50% to afford two multivalent sialyllactose derivatives $LPG(6'\text{-}SLN_3)_{0.50}$ and $LPG(6'\text{-}SLamide)_{0.50}$ with different linkages (Scheme 1).

Modified Functionalization of Scaffold Broadens Anti-Influenza Activity. Both the multivalent 6′-SL compounds $LPG_{10}(6'\text{-}SLN_3)_{0.50}$ and $LPG_{10}(6'\text{-}SLamide)_{0.50}$ were screened for the inhibition of different IAV strains using the same setup displayed in Figure 1. Interestingly, $LPG_{10}(6'\text{-}SLamide)_{0.50}$ reduced the titers of not only A/X31 virus (H3N2) with an IC_{50} value of 0.38 μM (Supporting Information Figure S1) but also the more recent A/Panama/2007/1999 (H3N2) strain as well as the pandemic H1N1 subtype virus A/Bayern/63/2009 by 2 to 3 orders of magnitude. The titer of the recent A/Bremen/5/2017 virus was also reduced, solidifying $LPG_{10}(6'\text{-}SLamide)_{0.50}$'s potential for broad antiviral activity (Figure 7). Interestingly, the analogue compound prepared with a SLN_3 derivate, that is, $LPG_{10}(6'\text{-}SLN_3)_{0.50}$, was not able to substantially reduce virus titers, indicating that the linkage of SL to the PG is critical for the inhibitory efficacy of the compounds. Unfunctionalized control compounds did not reduce virus titers (Figure 7). These data suggest that linkage, functionalization, or scaffold alone is not pivotal, but that a certain combination of a particular linkage with a particular scaffold is essential for

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Scheme 1. Synthesis of 6′-Sialyllalctose Derivatives and Their Conjugation with LPG Polymers[a]



[a]Reagents and conditions: (a) propargyl amine, RT, 48 h; (b) Ac_2O, dry MeOH, RT, 24 h; (c) NaN_3, 2-chloro-1,3-dimethylimidazolinium chloride, DIPEA, H_2O, 0 °C, 1 h; and (d) $CuSO_4{\cdot}5H_2O$, sodium ascorbate, DMF/H_2O, 50 °C, 48 h.

antiviral activity. The exact configuration of the compound is key to an effective and broad inhibition.

DISCUSSION AND CONCLUSIONS

Previous analyses showed that SA-functionalized PGs can be synthesized with different flexibilities and degrees of functionalization, but retain the common capacity of inhibiting the commonly used lab strain influenza A/X31 virus.[45] However, the present study revealed that SA-functionalized PGs with different scaffolds failed to inhibit a broader spectrum of IAV strains, presumably due to structural variations in the viral HA known be part of intra-subtypic evolution and inter-subtypic differences. To elucidate the molecular basis for this limitation in more detail, we selected, by serial passaging, resistance to prototypic $LPG_{10}SA_{0.40}$, a flexible linear PG with 40% SA density. Resistance developed readily and was conferred by amino acid changes in the HA head region altering the receptor binding profile and also the affinity to SA. Moreover, the identified mutations also conferred cross-resistance to other PGs with different degrees of functionalization or flexibility. Therefore, we hypothesized that a change in the functionalized groups would help to broaden the spectrum of susceptible virus strains and possibly the propensity to acquire resistance mutations. Glycan array analysis revealed that sialyllactose was a receptor shared by WT virus and mutant viruses. Hence, we synthesized PGs functionalized with sialyllactose (SL) in 6′-configuration, a typical receptor of human influenza viruses.[78] Indeed, an LPG functionalized with SL with an amide linker (LPG_{10}-(6′-SLamide)$_{0.50}$) showed an extended spectrum of inhibited virus strains and may be considered a promising candidate for broad inhibition of different influenza viruses.

To investigate the molecular determinants of susceptibility to SA-functionalized PGs, three independent variants of the influenza A/X31 virus (H3N2) were selected by serial passaging in the presence of $LPG_{10}SA_{0.40}$, which carried either the HA mutations G135E or K140N or P99F/G144D, respectively (H3 numbering) (Table 1). These mutations map in or close to the RBS (Figure 3C),[10] which was not unexpected, because $LPG_{10}SA_{0.40}$ is a competitive inhibitor of

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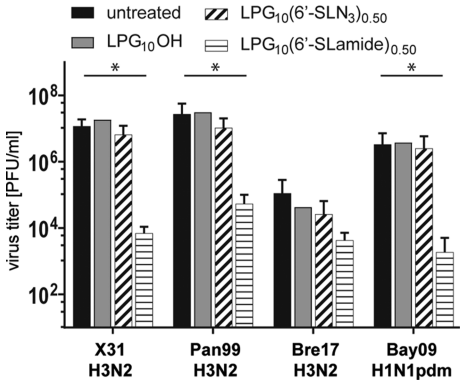


Figure 7. Therapeutic inhibition of virus infection using 6′-SL conjugated compounds. MDCK II cells were infected with A/X31 (X31), A/Panama/2007/1999 (Pan99), A/Bremen/5/2017 (Bre17), or A/Bayern/63/2009pdm (Bay09) at MOI 0.01 and treated with 5 μM [PG] of the indicated compound for 24 h. Each column represents the mean of three independent experiments. Error bars depict the SD. *P*-values were calculated using the Kruskal−Wallis test with Dunn's correction for multiple comparisons. * indicates a *p*-value < 0.05.

HA receptor binding.[15,45] For instance, G135 of influenza virus A/X31 is in direct contact with SA via a hydrogen bond.[10] Thyagarajan and Bloom employed deep mutational scanning to reveal that positions 135, 140, and 144 of H3 subtype HA have an overall high mutational tolerance with no clear preference for a specific amino acid.[88] This may be a reason why this region has evolved as highly variable antigenic site A that is under selection pressure by antibodies.[89−91] G135E and G144D have occurred naturally during evolution of H3N2 subtype influenza viruses as part of transition between antigenic clusters, and changes at K140 have occurred in circulating strains as well.[89] Thus, the genetic barrier for influenza A/X31 virus to develop resistance by adaptive mutations in these positions appears to be relatively low. The early development of adaptive mutations observed here (Figure 2, Table 1) is comparable to the development of resistance in influenza viruses to other substances, like amantadines, which developed within two to three passages,[92] resistance against Oseltamivir in passage three,[93] and resistance against Zanamivir as early as passage one.[72]

Upon serial passaging with Oseltamivir, only variants harboring mutations in HA and not NA were selected here. It is well established that HA mutations decreasing receptor affinity can confer resistance to NA inhibitors without changes in NA activity *in vitro*.[71,72] G135E in H3 subtype HA found in the present study in particular has been previously described to confer resistance to Oseltamivir.[93]

Because the resistance mutations occurred around the RBS, we assessed putative changes in receptor binding by analysis with two glycan arrays (Figure 5): WT virus bound with high binding intensity to established receptors of A/X31 virus attachment.[85] In contrast, L1 and L2 bound only to some of the established receptors, with binding intensities reduced by 2 orders of magnitude, whereas no binding was detected for L3. Interestingly, the mutant viruses bound several distinct alternative glycans. The entropic penalty of 6′-Neu5Ac binding to HA is rather large because it has one rotatable bond more than 3′-Neu5Ac, allowing for multiple conformations and also more contact points with HA.[94] In order to overcome the entropic penalty and facilitate binding, a higher affinity is required.[95] The reduced binding of mutant viruses to 6′-Neu5Ac might be due to the reduced affinity not being able to outweigh the entropic penalty incurred by the conformational restriction of the 6′-Neu5Ac receptor upon binding. Several groups have emphasized that the shape of the receptor is the determining factor in binding and not the SA itself.[78,94,96] Neu5Acα2−8Neu5Acα substructures, Neu5Gc, and even terminal galactose or GlcNAc have been identified as receptors for influenza viruses of different host species including human-origin influenza viruses.[78] The reduced affinity for the prototypical 6′-Neu5Ac receptor could lead the mutant viruses to bind these alternative receptors. It is entirely possible that we did not detect additional receptors of the mutant viruses here because the arrays display a limited fraction of short glycans present on tissue culture cells[82−84] or in respiratory tissues.[9]

Despite a reduction of affinity for the receptors used by WT virus, the mutant viruses displayed growth curves similar to WT virus (Figure 3A). There is a broad variety of glycans present in the lung[9] and also on MDCK II cells.[82−84] Hence, changes in receptor specificity or affinity do not necessarily have a negative impact on viral fitness or infectivity, as the IAV can apparently switch to alternatively preferred determinants.[97] It appears that $LPG_{10}SA_{0.70}$ and $dPG_{10}SA_{0.15}$ are too similar to $LPG_{10}SA_{0.40}$ and the difference in SA density or flexibility is not sufficient to overcome the altered receptor preference of the mutant viruses, resulting in cross-resistance (Figure 3B).

Interestingly, SA binding of the mutant viruses was reduced, but not abolished, as indicated by AFM measurements (Figure 4), glycan array analysis (Figure 5), and the sustained inhibition of mutant virus propagation by phage capsid Qβ[sia1][43] (Figure 3B). Phage capsids are spatially designed to match the three RBS on HA trimers.[43] Therefore, these reagents can act as very robust anti-influenza substances that are less susceptible to subtle differences in HA. This may be due to a higher avidity, resulting in successful inhibition despite a reduced affinity for SA in the mutant HA. Moreover, Cy3-labeled $LPG_{10}SA_{0.40}$ was retained in cells infected with WT and mutant viruses, but selectively inhibited only propagation of WT virus (Figure 6). The pattern observed for the mutant viruses is similar to that of A/Panama/2007/1999 (H3N2) virus, whereas no compound retention or inhibition was observed for the H1N1 strain A/PR8/34 (Figure 1C). According to their amino acid sequence, H1 belongs to group 1 HAs, whereas H3 belongs to group 2 HAs, indicating their high divergence. In fact, the same sialylated receptor adopts a slightly different conformation when bound to the RBS of H3 compared to H1.[98] Furthermore, the different glycosylation patterns of H3 and H1 on the HA head might contribute to differences in inhibitor binding.[99] Therefore, multivalent sialosides binding H3 might bind H1 with lower affinity or not at all, as also indicated by the different effectivity of multivalent sialosides toward H3N2 and H1N1 viruses often observed.[33,43] Taken together, we have three cases: (1) viruses being bound and inhibited, (2) viruses being bound but not inhibited, and (3) viruses being not bound and not inhibited. While the first and the third case appear rather obvious, the second case is especially interesting because it implies that binding alone is not sufficient for inhibition of virus propagation. Rather, a certain affinity of a compound to HA is required for inhibition of virus propagation. If the affinity is too low as is the case for the mutant viruses and SA-functionalized PGs, the compound may

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be capable of binding HA but may be easily displaced by cellular receptors and thus not impact virus propagation.

In an effort to refine the PGs for efficient and broad inhibition of IAV propagation, we focused on the sugar functionalization and linkage. SL has been successfully incorporated in multivalent compounds inhibiting IAV propagation[32,33,47,49,100] and has even been earlier suggested to be able to inhibit all human IAV.[63,64] Multivalent PAMAM conjugates that were functionalized with SA were subtype and even strain-specific,[31] but further developed PAMAM constructs incorporate SL and, despite requiring a compound concentration in the higher μM range, successfully inhibit a variety of influenza viruses.[32,33] Hence, our next step was the synthesis of PGs carrying SL instead of SA alone. The linker or spacer proved to be an important factor in effective inhibition using PAMAM constructs,[30] PEG and DNA scaffolds,[101] phage capsids,[43] glycocopolymers,[47] diverse other scaffolds,[49] and molecular modeling.[48] Therefore, 6′-SL derivatives with amide or azide groups at the anomeric position were synthesized to obtain two versions with different linkers. In fact, the SL-PG with the amide group was able to inhibit the propagation of diverse IAV strains by several orders of magnitude albeit the analogue compound with the azide group showing little inhibitory effect (Figure 7). These findings suggest that the nature of the linker is crucial.

Earlier developments of sialylated compounds have emphasized that it is a challenge to achieve antiviral activity covering a broad spectrum of IAV strains or even subtypes.[23,30,31,47,61] We were able to address this challenge by PGs functionalized with SL and the right linker. PGs are a promising scaffold system because besides being biocompatible, their flexible and easily modifiable nature allows them to accommodate for many modifications. Future work could combine several sugars on one PG, for example, the top six glycans identified by Gulati et al. which were bound most often by the 45 H3N2 viruses isolated between 1968 and 2012 with one virus to represent each year.[97] This approach could strengthen broad activity and raise the threshold for resistance development by limiting the options for altering the receptor binding profile. In order to synthesize inhibitors against HA receptor binding that effectively inhibit a wide variety of influenza virus strains, extensive knowledge of the receptor interaction of HA and the effect of differences on this interaction is key. Hence, this work provides novel insights into the processes accompanying the acquisition of mutations in the RBS of the viral HA under selection pressure and how multivalent SA conjugates can serve as broad influenza virus inhibitors.

EXPERIMENTAL SECTION

Materials for Synthesis and Characterization. All reagents and solvents were purchased from commercial suppliers and used without further purification. Reactions requiring dry or oxygen-free conditions were carried out under argon in Schlenk glassware. ^{1}H spectra were recorded on Bruker AMX 500 (500 MHz) and Delta Joel Eclipse 700 (700 MHz) spectrometers at 25 °C and calibrated by using the deuterated solvent peak. Infrared (IR) spectra were recorded with a Nicolet AVATAR 320 FT-IR 5 SXC (Thermo Fisher Scientific, Waltham, MA, USA) with a DTGS detector from 4000 to 650 cm^{-1}. A TSQ 7000 (Finnigan Mat) instrument was used for ESI measurements, and a JEOL JMS-SX-102A spectrometer was used for the high-resolution mass spectra. Molecular weight distribution of $LPG_{10}OH$ was determined by means of GPC coupled to a refractive index detector (RI) for obtaining the complete distribution (M_n, M_p, M_w, dispersity) under highly diluted conditions of 5 mg/mL with a GPC consisting of an Agilent 1100 solvent delivery system with pump, manual injector, and an Agilent differential refractometer. Three 30 cm columns (PPS: Polymer Standards Service GmbH, Germany; Suprema 100, 1000, 3000 Å with 5 and 10 mm particle size) were used to separate aqueous polymer samples using water with 0.1 N $NaNO_3$ as the mobile phase at a flow rate of 1 mL/min. The columns were operated at room temperature (RT) with the RI detector at 50 °C. The calibration was performed by using certified standards pullulan (linear) and dextran (branched) from PSS. Water of Millipore quality was used in all experiments. Dialysis tubing of molecular weight cut-off 8 kDa and 500 Da purchased from Spectrum Labs was used to purify the final polymers and sialyllactose derivatives by dialysis. All tested compounds are >95% pure by ^{1}H NMR.

6′-Sialyllactosylamide (6′-SLamide). 6′-SLamide was prepared in two consecutive steps. The mixture of α-2,6-sialyllactose (1.0 g, 1.58 mmol) and neat propargyl amine (3.0 mL, 47.35 mmol) was stirred at RT for 48 h. After completion of reaction as monitored by thin-layer chromatography (TLC), the reaction mixture was concentrated under reduced pressure and the crude obtained was dissolved in dry methanol (25 mL) and acetic anhydride (3.0 mL, 31.74 mmol) and stirred again at RT. After 24 h, the reaction mixture was concentrated again and co-evaporated with toluene three to four times. The obtained residue was first dialyzed using the 100−500 MWCO dialysis membrane in water for 2 days. The lyophilized material was purified by preparative HPLC using MeOH−CH_3CN with 1% H_2O as a solvent system to afford the desired pure product as a white solid (1.0 g, 92% yield). ^{1}H NMR (500 MHz, D_2O): δ 5.53 (d, J = 9.0 Hz, 0.3H, α isomer), 5.07 (d, J = 8.8 Hz, 0.7H, β isomer), 4.42−4.40 (m, 1H), 4.29−4.13 (m, 2H), 3.98−3.49 (m, 21H), 2.68 (dd, J = 12.4, 4.5 Hz, 1H), 2.22, 2.29 (s, 3H, two isomers), 1.99 (s, 3H), 1.72 (t, J = 12.2 Hz, 1H). ^{13}C NMR (125 MHz, CD_3OD): δ 176.1, 175.1, 174.9, 173.7, 103.4, 100.4, 86.5, 81.9, 79.2, 76.8, 75.4, 73.9, 72.7, 72.5, 72.0, 71.7, 71.0, 69.9, 68.8, 68.5, 63.8, 62.8, 60.4, 51.8, 48.8, 40.4, 33.0, 30.2, 22.2, 21.8, 21.4. HRMS (ESI): m/z calcd for $C_{28}H_{44}N_2O_{19}$, 735.2436 [M + Na$^+$]; found, 735.2425. IR (neat, cm^{-1}): ν_{max} 3276, 2932, 2884, 1612, 1378, 1325, 1072, 1034. Spectral data are reported in the Supporting Information.

LPG_{10}(6′-SLN_3)$_{0.50}$. The 10 kDa-based $LPG_{10}(propargyl)_{0.50}$ was prepared starting from $LPG_{10}OH$[102] (M_n = 10.1 kDa, D = 1.29), as reported earlier.[103] To the $LPG_{10}(propargyl)_{0.50}$ (100 mg, 7.97 μmol of polymer, ~67 propargyl groups per polymer) in dimethylformamide (DMF) (10 mL), 6′-SLN_3 (387 mg, 0.59 mmol, 1.1 equiv per propargyl unit) was added followed by addition of a mixture of $CuSO_4 \cdot 5H_2O$ (26 mg, 0.107 mmol, 0.2 equiv per alkyne unit) and sodium ascorbate (42 mg, 0.214 mmol, 0.4 equiv per propargyl unit) dissolved in a minimum amount of water. The reaction mixture was degassed with argon and stirred at 50 °C for 24 h, followed by evaporation of solvents under reduced pressure. Collected residue was dissolved in water and dialyzed against water and ethylenediaminetetraacetic acid (EDTA) for 3 days using the 8 kDa MWCO dialysis membrane. The pure product was obtained as a white solid (371 mg, 82% yield). The DF was quantified by ^{1}H NMR as 50%. ^{1}H NMR (500 MHz, D_2O): δ 8.55 (br s, 1H), 5.88 (br s, 1H), 4.52 (br s, 1H), 3.97−3.59 (m, 28H), 2.74 (br s, 1H), 2.07−1.87 (m, 4H). ^{13}C NMR (125 MHz, CD_3OD): δ 175.3, 175.2, 103.5, 103.1, 87.8, 87.7, 79.9, 78.9, 77.8, 77.6, 75.6, 75.0, 74.8, 73.9, 72.8, 72.6, 72.1, 72.0, 71.0, 69.2, 68.8, 68.7, 68.5, 63.8, 62.9, 61.3, 61.0, 60.9, 60.1, 52.0, 40.3, 22.3. MW (kDa) = 55 kDa (^{1}HNMR). IR (neat, cm^{-1}): ν_{max} 3348, 2927, 2884, 1617, 1378, 1320, 1067, 1043. Spectral data are reported in the Supporting Information.

LPG_{10}(6′-SLamide)$_{0.50}$. The preparation of 10 kDa-based LPG-N_3 with DF = 60% was prepared as reported earlier.[104] To a stirred solution of LPG-N_3 (50% azide-functionalized 10 kDa) [100 mg, 8.62 μmol of polymer, 67 (50%) N_3 groups to be functionalized] in DMF (10 mL), 6′-SLamide [462 mg, 0.63 mmol, 1.1 equiv per azide group (50% functionalization)] was added, followed by addition of a mixture of $CuSO_4 \cdot 5H_2O$ (27 mg, 0.116 mmol, 0.2 equiv per azide group) and sodium ascorbate (43 mg, 0.232 mmol, 0.4 equiv per azide group) dissolved in a minimum amount of water. The reaction mixture was

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degassed with argon and stirred at 50 °C for 48 h, followed by evaporation of solvents under reduced pressure. Collected residue was dissolved in water and dialyzed against saturated EDTA solution and then water for 3 days. The dialyzed solution was lyophilized to obtain the desired product as a white solid (420 mg, 86% yield). The DF was quantified by ^{1}HNMR as 50%. ^{1}H NMR (500 MHz, D_2O): δ 8.05 (br s, 1H), 4.44 (br s, 1H), 3.95−3.57 (m, 32H), 2.72 (br s, 1H), 2.13−1.76 (m, 7H). ^{13}C NMR (125 MHz, D_2O): δ 175.1, 173.5, 103.4, 86.9, 86.5, 83.3, 79.8, 79.3, 79.2, 77.9, 76.8, 75.4, 73.9, 72.7, 72.5, 72.0, 71.0, 70.1, 69.1, 68.8, 68.7, 68.5, 63.7, 62.8, 60.9, 52.0, 40.3, 22.2, 21.3, 20.4. MW (kDa) = 58 kDa (^{1}H NMR). IR (neat, cm^{-1}): ν_{max} 3283, 2928, 2884, 2102, 1616, 1374, 1238, 1035. Spectral data are reported in the Supporting Information.

Cells and Viruses. MDCK II (Madin−Darby canine kidney epithelial) cells (ATCC) were maintained at 37 °C and 5% CO_2 in minimum Eagle's essential medium (MEM) supplemented with 10% fetal bovine serum, 2 mM L-glutamine, 100 μg/mL streptomycin, and 100 units/mL penicillin. Infected cells were maintained in infection medium (MEM supplemented with 0.2% bovine albumin, 2 mM L-glutamine, 100 mg/mL streptomycin, 100 units/mL penicillin and 1 μg/mL TPCK-Trypsin). A/Panama/2007/1999 (H3N2), A/Bremen/5/2007 (H3N2), and A/Bayern/63/2009 (H3N2pdm) were harvested from a cell culture supernatant of MDCK II cells. A/PR/8/34 (H1N1) and X31 virus (A/PR/8/34 (H1N1) reassorted with HA and NA from A/Aichi/2/68 (H3N2)) were harvested from allantoic fluid of embryonated chicken eggs. Stock viruses were centrifuged at 3000 rpm for 5 min for clarification.

Inhibition of Viral Replication. MDCK II cells at 90% confluency were infected with the indicated influenza virus at a multiplicity of 0.01 for 45 min at RT after which the input virus was removed. The cells were then incubated in infection medium supplemented with the indicated compound at the indicated concentration for 24 h. 100 μM [SA] or the equivalent amount of unfunctionalized PG was used for growth analysis. For IC_{50} determination, a fivefold dilution series of compound was applied. Virus titers in the supernatant were determined by plaque titration on MDCK II cells, and the results are given as plaque forming units per mL (PFU/mL). IC_{50} values were calculated using GraphPad Prism. All data represent at least three independent experiments in duplicates.

Serial Passaging. MDCK II cells at 90% confluency were infected with IAV A/X31 at a multiplicity of 0.01 for 45 min at RT, after which the input virus was removed. The cells were then incubated in infection medium supplemented with $LPG_{10}SA_{0.40}$ at increasing concentrations (from passage 1 to passage 10: 0.1, 0.1, 0.2, 0.3, 0.8, 2, 4, 20, 100, and 100 μM [SA]) or left untreated for 24 h. After plaque titration, supernatants were used to infect the next passage. The wild-type virus and supernatants of passages 4, 7, and 10 were processed for NGS.

Next Generation Sequencing. Total RNA was extracted from cell culture supernatants of the serial passaging experiment using the ReliaPrep RNA Cell Miniprep System (Promega) as per manufacturers' instructions. Viral RNA was amplified using the Super Script III One-Step RT-PCR System with Platinum Taq High Fidelity (Invitrogen by Life Technologies) as described previously.[105] The PCR product was purified using the MSB SPIN PCRapace Kit (Stratec) as per the manufacturers' protocol 1a. Purified samples were sequenced using an Illumina MiSeq in the Robert Koch-Institute in-house sequencing facility.

Plaque-Picking of Mutant Viruses. MDCK II cells at 90% confluency were infected with the supernatant of passage 10 of the serial passaging experiment at a multiplicity of 0.01 for 45 min at RT, after which the input virus was removed. The cells were then incubated in agar medium [double-strength MEM supplemented with 0.2% bovine albumin, 1% (diethylamino)ethyl dextran, 5% sodium bicarbonate, 1 μg/mL trypsin-TPCK, and 2% Oxoid agar]. After 48 h, single plaques were picked, dissolved in PBS, and used to infect fresh flasks of MDCK II cells. Input virus was removed after 45 min, and cells were incubated in infection medium for 72 h. Supernatants were centrifuged at 3500 rpm for 5 min to remove cell debris.

Sanger Sequencing. RNA was extracted using the QIAamp MinElute Virus Spin kit (Qiagen) according to manufacturers' instructions. cDNA was generated by combining the extracted RNA with primers described before that are capable of amplifying all eight influenza virus genomic RNA segments[105] at 65 °C for 5 min and then adding the RevertAid Reverse Transcriptase System (Thermo Scientific) for incubation at 42 °C for 50 min and 70 °C for 10 min. cDNA was amplified using 150 nM of the following primers: 3′-CGTAGAATTCCATCATTGCTTTGAGCTACATTTT-5′, 3′-CGTATCTAGATGCACCTAATGTTGCCTCTCT-5′ and 3′-CAC-GAACCAAGAACAAACCA-5′ and the Taq DNA Polymerase system (Invitrogen by Life Technologies). PCR was performed for 2 min at 95 °C, followed by 35 cycles of melting at 95 °C for 30 s, annealing at 59 °C for 30 s, extension at 72 °C for 2 min, and a final extension step at 72 °C for 5 min. The PCR product was purified using the MSB SPIN PCRapace Kit (Stratec) as per the manufacturers' protocol 1a. The sequencing reaction was performed using the BigDye Terminator v3.1 Cycle Sequencing Kit (Applied Biosystems) using 1 μL of BigDye, 1.5 μL of 5× Buffer, 6 μL of deionized water, 5 pmol of primer, and 15−20 ng of cDNA template. The temperature profile was 96 °C for 2 min followed by 25 cycles of 96 °C for 10 s, 59 °C for 5 s, and 60 °C for 2 min. The sequencing was conducted using the dideoxy-chain termination method in an automated DNA sequencer at the Robert Koch Institute in-house sequencing facility.

Glycan Array Analysis. All virus stocks analyzed on the glycan arrays were grown on MDCK II cells. Binding specificity was analyzed on the Glycan Array 100 (RayBiotech) with the sandwich-based approach depending on biotinylation of the primary antibody. Briefly, 400 μL of virus solution (125 μg/mL of virus in binding buffer) was incubated on the array at 4 °C o.N. Slides were washed (3 × 5 min) with PBS containing 0.1% Tween 20 at RT. Virus particles were then fixed by incubation with 2.5% formaldehyde for 15 min at RT, followed by permeabilization using PBS containing 0.2% TritonX100 for 10 min at RT. Slides were then washed and blocked with sample diluent for 3 h at RT. The remaining procedure was conducted as per the manufacturers' instructions. For detection, a primary monoclonal antibody against the viral NP protein (1:1000, clone AA5H, AbD Serotec) was biotinylated using the LYNX Rapid Plus Biotin (Type 1) Antibody Conjugation Kit (Bio-Rad) according to manufacturers' instructions. The low-density array was printed at the Max Planck Institute of Colloids and Interfaces and analyzed as described previously.[85] Of each virus, 2 × 10^6 PFU were loaded on the low-density array, which amounted to the same HA titer. Fluorescence images were obtained using a GenePix microarray scanner (Molecular Devices) and analyzed with the GenePix pro 7.2 software.

Confocal Microscopy. MDCK II cells grown to 90% confluency on sterile cover slips in 24-well plates were infected with the indicated virus at an MOI of 0.1 for 45 min at RT, after which the input virus was removed. The cells were then incubated in infection medium supplemented with the indicated Cy3-labeled compound at the indicated concentration for 24 h. Cells were washed, fixed with 2.5% paraformaldehyde, and permeabilized with 0.2% TritonX100. A primary rabbit monoclonal antibody against the viral NS1 (1:1000 in PBS with 3% BSA) and a primary mouse monoclonal antibody against cellular Na^+/K^+-ATPase (1:100 in PBS with 3% BSA) were applied for 2 h. Residual antibody was washed off followed by 1 h incubation with secondary antibodies anti-rabbit coupled Alexa488 and anti-mouse coupled Alexa647. DAPI was used to stain cellular DNA. Images were taken using a Zeiss LSM 780 Axio Observer and the ZEN black software.

AFM Experiments. Experiments were done with a Nanowizard 3 (Bruker) using the force spectroscopy mode. AFM cantilevers of the type MLCT-C with a nominal spring constant of 0.01 N/m were used. Cantilevers were calibrated individually at the beginning of each experiment. Each force−distance curve was recorded on a different spot. Recordings were done in a series of up to 400 force−distance curves on grids of 20 × 20 points over an area of up to (20 × 20) μm2 at 0.5, 1.0, 3.0, 5.0, and 10.0 μm/s retraction speed. Single virions were bound to the AFM cantilever using an established protocol from JKU Linz.[106] We used a heterobifunctional 3,4 kDa Acetal-PEG-NHS

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linker purchased from Creative PEGWorks (Research Triangle Park, United States of America). First, the AFM Cantilever was amino-functionalized, followed by binding of the linker via the NHS group. Before single virions of ultracentrifuge grade were bound to the linker, the acetal group was activated to an aldehyde group by incubation in 1% citric acid. Amino-functionalized as well as linker-bound AFM cantilevers were stored for several days up to several weeks. AFM cantilevers with bound single virions were prepared freshly on the first day of a 2-day experiment series. Each AFM cantilever was used only for 1 day. Freshly prepared AFM cantilevers with bound virus were stored in phosphate-buffered saline without magnesium and calcium (PBS--) at 4 °C. SA displaying receptors were presented on manufactured glass slides from MicroSurfaces (Englewood, Unites States of America). The glass slides are covered by a PEG layer featuring streptavidin molecules. Glass slides were prepared freshly before force experiments by incubation with a 0.5 mg/mL 6′-sialyllactose-biotin solution (Dextra Laboratories, Bristol, Great Britain).

Single Virus Force Spectroscopy Data Analysis. Recorded force−distance curves were analyzed with JPK Data Processing software and OriginLab 2017. The curves were baseline-corrected, and the contact point was determined. For the following steps, only curves with a rupture event indicated by a sawtooth shaped peak in the retract were considered. Peaks were fitted using the freely jointed model, and the rupture forces and critical loading rates were extracted. Extracted parameters were plotted in histograms, whereas only curves were considered with a contour length between 12 and 35 nm to account for the PEG chain length. Most probable rupture forces and loading rates were obtained by multimodal Gaussian fit and plotted against each other. The kinetic parameters dissociation rate and rupture length were obtained by applying the Bell−Evans model.[75]

$$F(r) = \frac{k_B T}{x_\beta} \ln\left(\frac{r \cdot x_\beta}{k_{off} \cdot k_B T}\right) \quad (1)$$

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jmedchem.1c00794.

Calculation of IC_{50} values, full list of glycans on the array, experimental data for Cy3-labeled unfunctionalized control compounds and additional glycan array data, synthetic details, NMRs, and GPC (PDF)

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

dPG, dendritic polyglycerol; GlcNAc, *N*-acetylglucosamine; HA, hemagglutinin; IAV, influenza A virus; LPG, linear polyglycerol; MOI, multiplicity of infection; Neu5AC, *N*-acetylneuraminic acid; Neu5Gc, *N*-glycolylneuraminic acid; o.N., overnight; PAMAM, polyamidoamine; PG, polyglycerol; RBS, receptor binding site; SA, sialic acid; SL, sialyllactose

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